

WASP-49b

R-1141

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_161208 · created 2026-07-31T00:05:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457730.8742 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.041
Transit depth (Rp/Rs)^2	1.69 +/- 1.05 [%]
Orbital Inclination (inc)	83.12 +/- 2.82
Airmass coefficient 1 (a1)	0.62 +/- 0.34
Airmass coefficient 2 (a2)	0.21 +/- 0.22
Scatter in the residuals of the lightcurve fit is	53.91 %
Best Comparison Star	#3 - [395, 64]
Optimal Aperture	2.05
Optimal Annulus	3.0
Transit Duration (day)	0.032 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.041 vs 0.116 (deviation 0.34σ)
Duration fit vs published	0.032 d vs 0.09 d (deviation 2.42σ)
Mid-time O–C	-5.0 min
Depth SNR	3.2
Residual scatter	53.91 %

Light curve

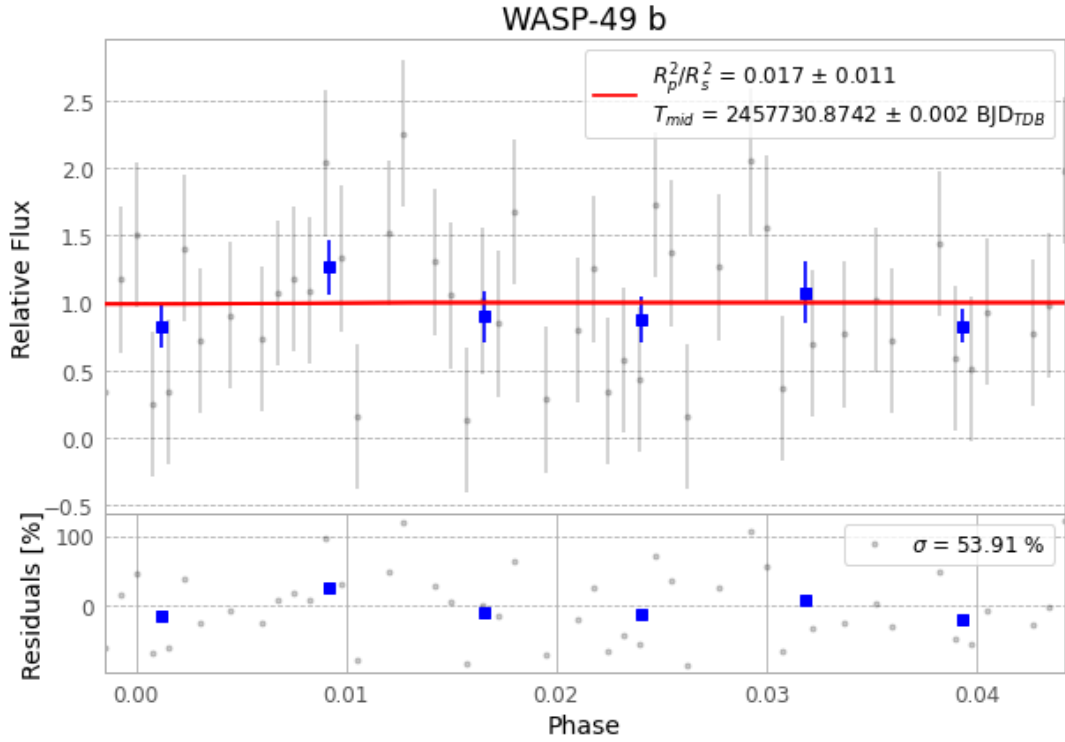


Figure 1 — FinalLightCurve_WASP-49 b_2016-12-08.png (SHA-256 99150727b88e2c12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 204], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c68eb9b181bd826d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), WASP-49161208083912.FITS → WASP-49161208115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-161208.log (dd144228b843...), FinalLightCurve_WASP-49 b_2016-12-08.png (99150727b88e...), FinalLightCurve_WASP-49 b_2016-12-08.csv (7f3c2d463fac...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:05Z.